

MTE 204: Numerical Methods

2.0 Roots of Equations

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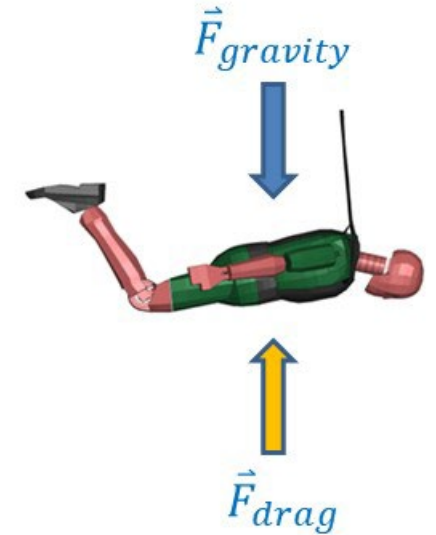
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2.0 Roots of Equations [Mathematical Models]

- Predict dependent variable as a function of independent variables
- Example (parachutist, terminal velocity)
 - $v = \frac{gm}{c} \left(1 - e^{-\frac{ct}{m}}\right)$
- **Explicit form**
 - Unknown, dependent variable isolated on left hand side of equation, known variables on right hand side
- $v = \frac{gm}{c} \left(1 - e^{-\frac{ct}{m}}\right)$ ← Given $v(t_1)$ at time t_1 solve for c
 - Cannot solve for c directly, implicit form
- From homogeneous equation
 - (RHS=0) and find values of c that satisfy the equation
$$\frac{gm}{c} \left(1 - e^{-\frac{ct}{m}}\right) - v(t_1) = 0$$
 - Then value of c is called root (or zero) of the equation

recall c = drag coefficient



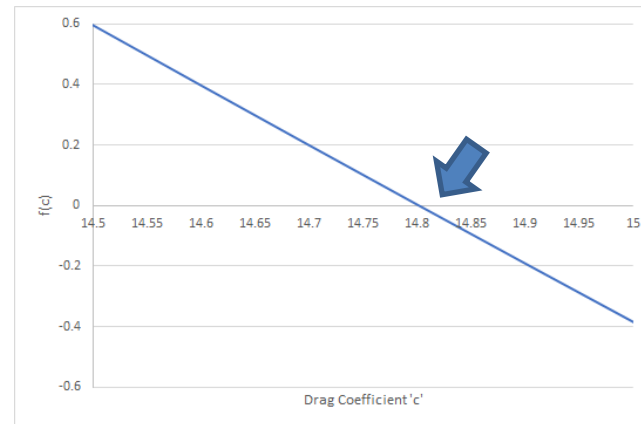
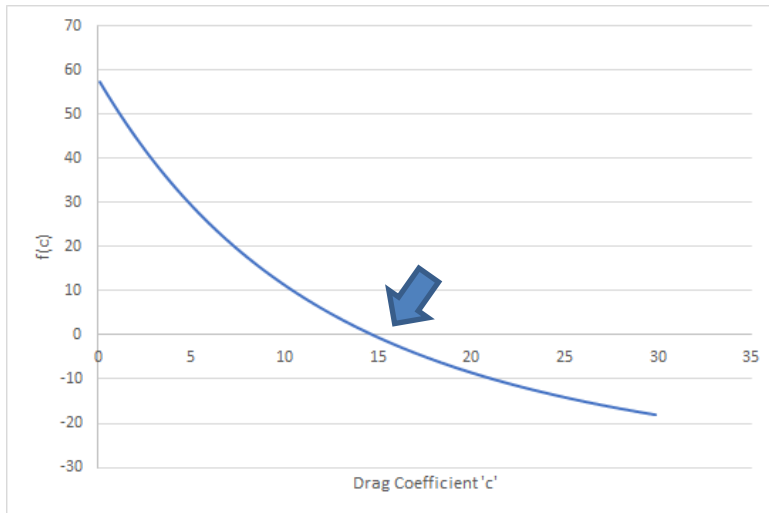
For example, we might be interested in the drag coefficient required to reach a specific velocity at a specific time, or for a given mass.

2.0 Roots of Equations [Mathematical Models]

[example 2.1]

$$\frac{gm}{c} \left(1 - e^{-\frac{ct}{m}}\right) - v(t_1) = 0$$

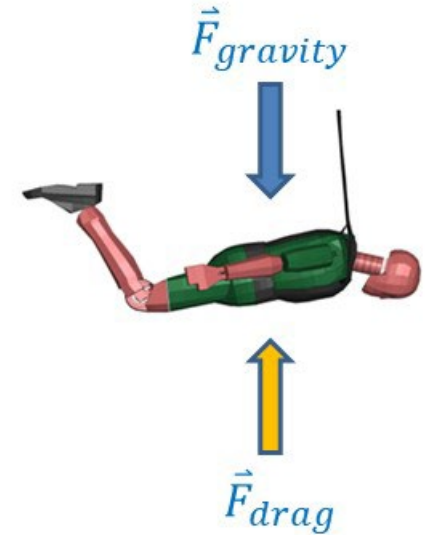
- Then value of c is called root (or zero) of the equation
- For $t = 10$ s, $g = 9.81$ m/s², $v = 40$ m/s, $m = 68.1$ kg



$$f(14.8) = 0.002213306 \approx 0 \text{ (close to zero!)}$$

Therefore, $c \approx 14.8$ to meet the required conditions.

recall c = drag coefficient



For example, we might be interested in the drag coefficient required to reach a specific velocity at a specific time, or for a given mass.

2.0 Roots of Equations [Mathematical Models]

[example 2.1]

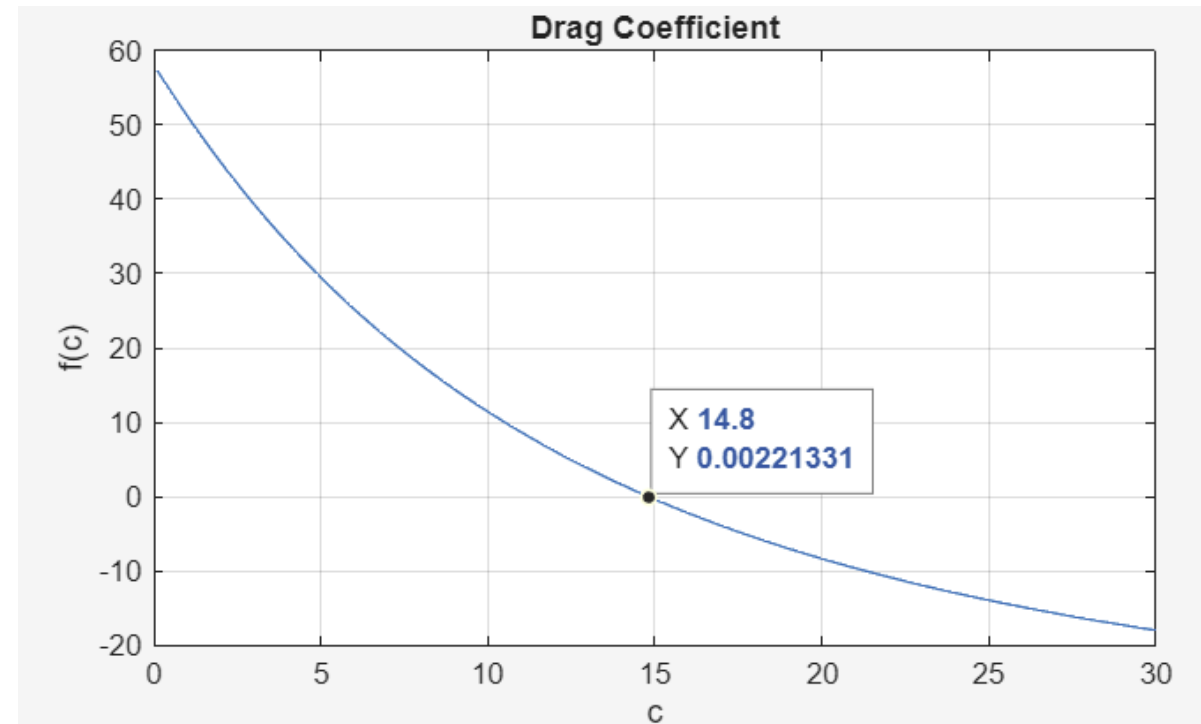
$$\frac{gm}{c} \left(1 - e^{-\frac{ct}{m}}\right) - v(t_1) = 0$$

- Then value of c is called root (or zero) of the equation
- For $t = 10$ s, $g = 9.81$ m/s², $v = 40$ m/s, $m = 68.1$ kg

In MatLab:

```
X=[0:.1:30];  
Y=(1.-exp(-X.*10.0./68.1)).*(X.^-1).*9.81.*68.1-40.;  
plot(X,Y), ylabel('f(c)'), xlabel('c'), title('Drag Coefficient'), grid  
on;
```

[s23.m]



2.0 Roots of Equations [Mathematical Models]

- Given second order polynomial equation

- $f(x) = ax^2 + bx + c = 0$

- Find roots with quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- Example

- $3x^2 + 4x + 1 = 0$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-4 \pm \sqrt{4^2 - 4(3)(1)}}{2(3)}$$

$$x = \frac{-4 \pm \sqrt{16 - 12}}{6}$$

$$x = \frac{-4 \pm \sqrt{4}}{6}$$

$$x = \frac{-4 \pm 2}{6}$$

$$x = -1, \quad x = -\frac{1}{3}$$

2.0 Roots of Equations [Mathematical Models]

- What about higher order or non-linear equations?

$x^2 + \sqrt{x} = 5$
 $\tan x = -x$ } Many methods available to find (1) real roots of (2)
algebraic equations and (3) transcendental equations

(1) Real Values

- Not complex values

- e.g. $x^2 + 1 = 0$

- $x^2 = -1$

- $x = \pm\sqrt{-1}$

- $x = \pm i$

(2) Algebraic equations

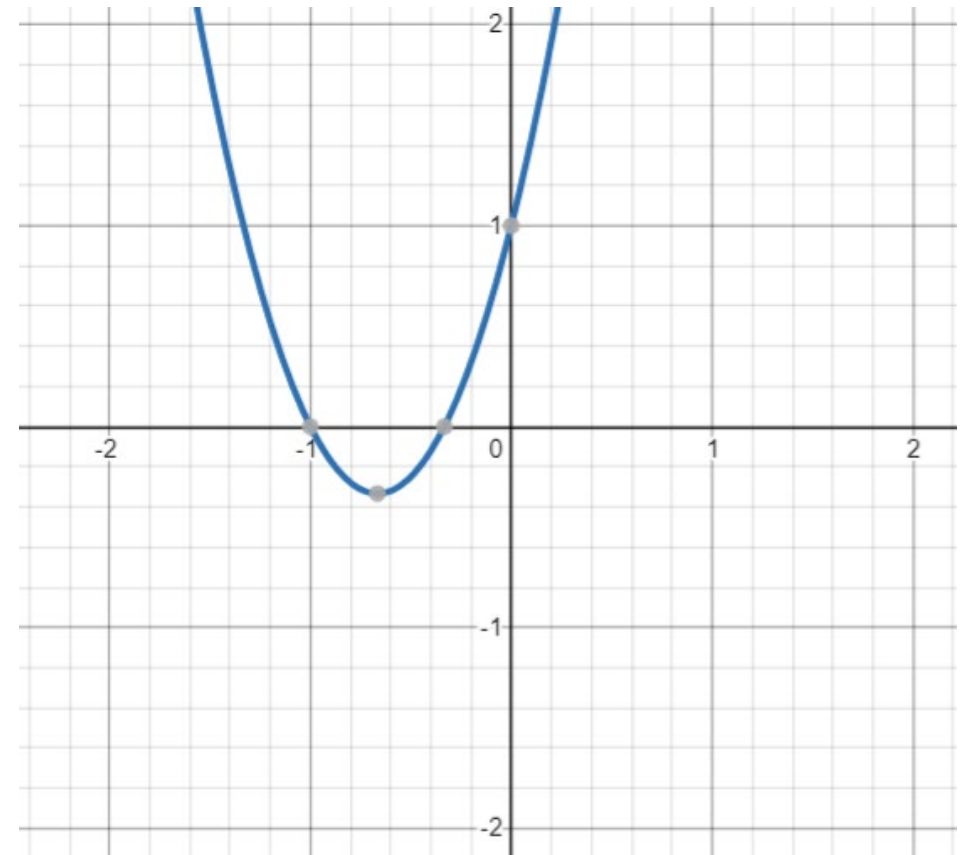
- Mostly polynomials

(3) Transcendental equations

- Includes trigonometric, logarithmic, exponential, other functions

2.1 Roots of Equations [Method #1 - Graphical]

- Make plot of function, observe where it crosses the x-axis
- Example
 - Find the roots of $f(x) = 3x^2 + 4x + 1$
 - i.e. where $f(x) = 0$ (crosses the horizontal axis)



2.1 Roots of Equations [Method #1 - Graphical]

- Make plot of function, observe where it crosses the x-axis
- Example
 - Find roots of $f(x) = 3x^2 + 4x + 1$ ← Parabola, opens upward

Minimum at $f'(x) = 0$

$$6x + 4 = 0$$

$$x = -\frac{2}{3}$$

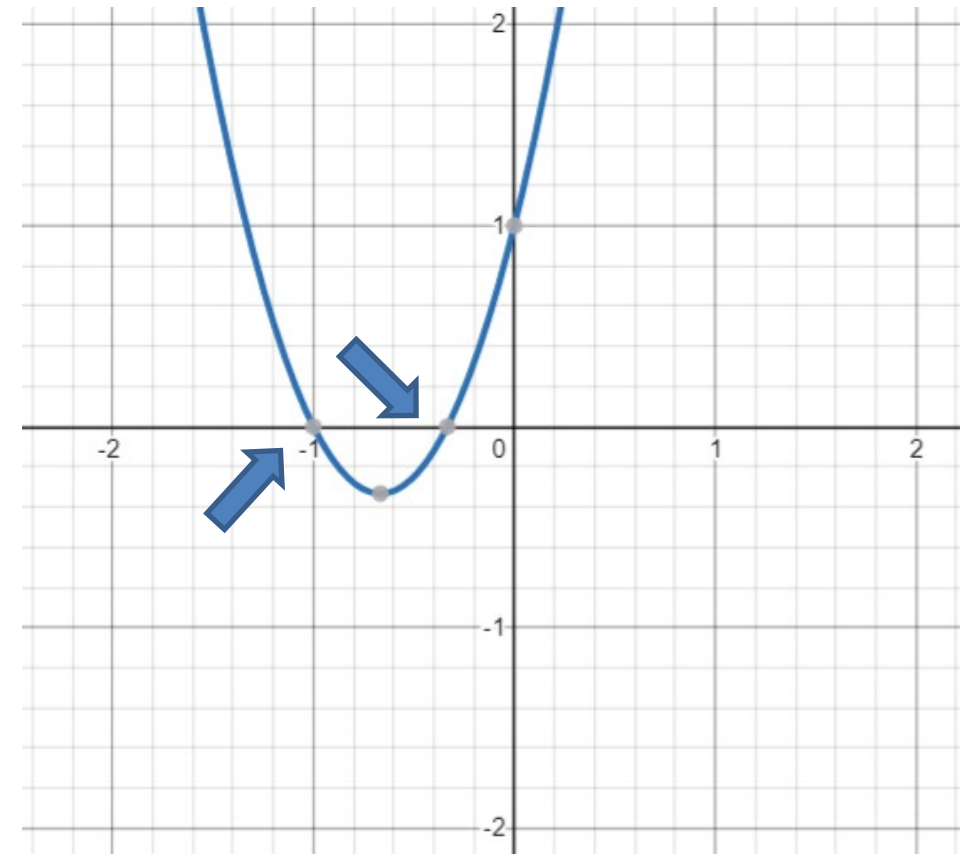
$$f\left(-\frac{2}{3}\right) = 3\left(-\frac{2}{3}\right)\left(-\frac{2}{3}\right) + 4\left(-\frac{2}{3}\right) + 1 = -\frac{1}{3}$$

Crosses the horizontal axis (by observation) at:

$$x = -1$$

$$x = -1/3$$

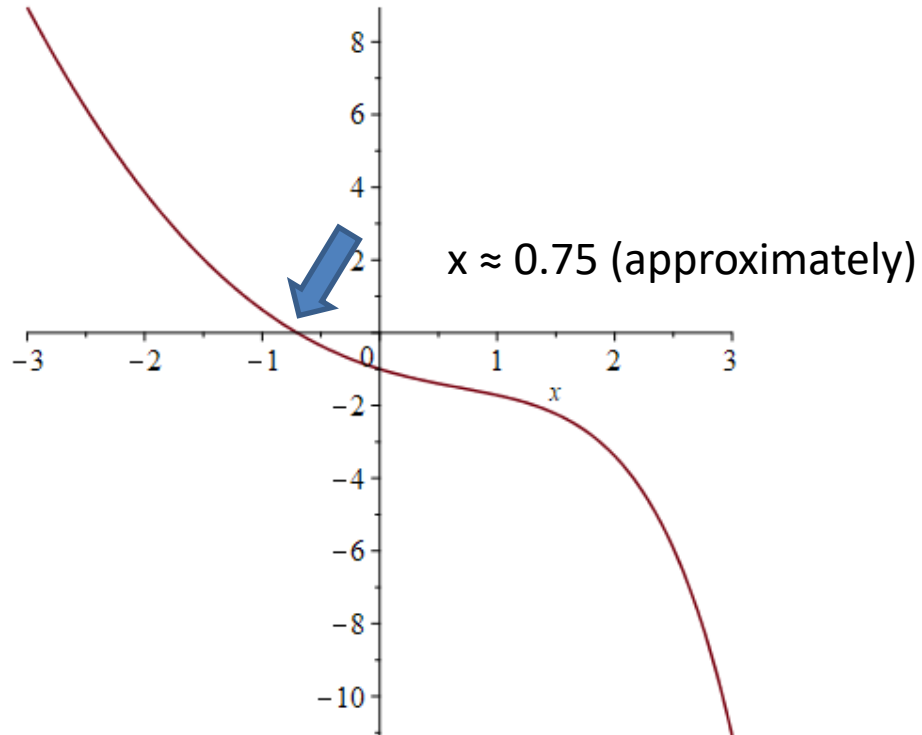
... or by using a graphical calculator



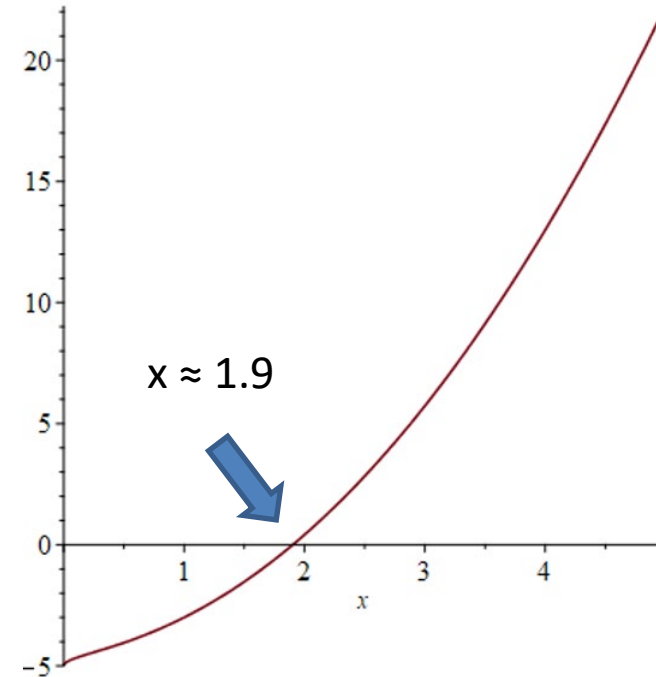
2.1 Roots of Equations [Method #1 - Graphical]

- Convenient to graph functions using math software, i.e. Matlab, Maple, Desmos etc.

$$f(x) = x^2 - e^x$$



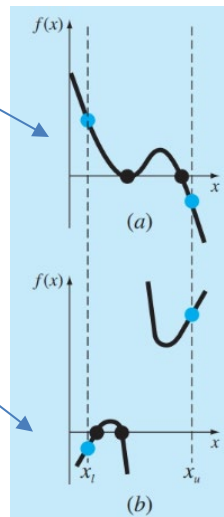
$$f(x) = x^2 + \sqrt{x} - 5$$



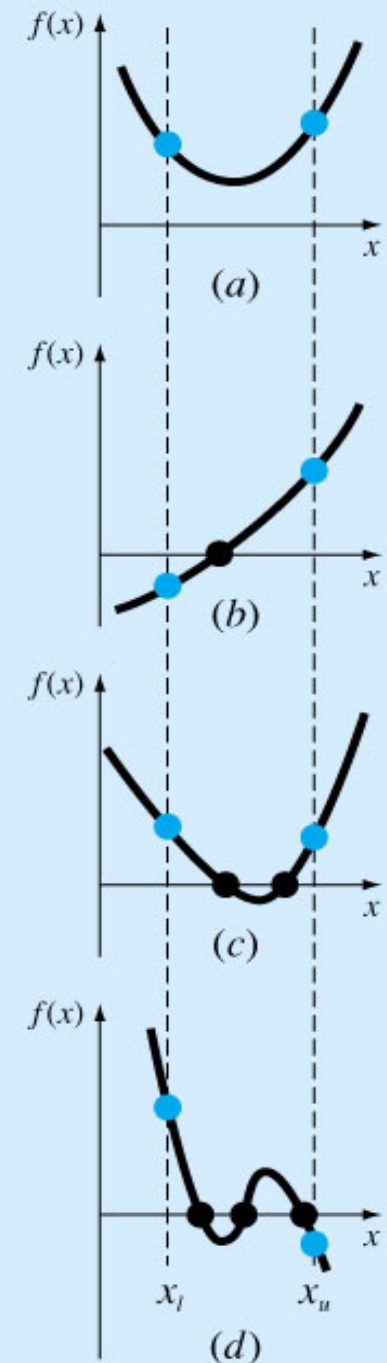
- Graphical methods can predict number of roots, approximate location of roots (starting point for analysis)

2.1 Roots of Equations [Method #1 - Graphical]

- General “rules of thumb” for root prediction based on graphical methods
 - If $f(x_i)$ and $f(x_{i+1})$ have the same sign, interval has no roots (a) or an even number of roots (c)
 - If $f(x_i)$ and $f(x_{i+1})$ have different signs, there is at least one root (b) or an odd number of roots (d)
 - Exceptions: $f(x)$ tangent to x-axis as minimum or maximum, or function is discontinuous



Even # of roots in both cases



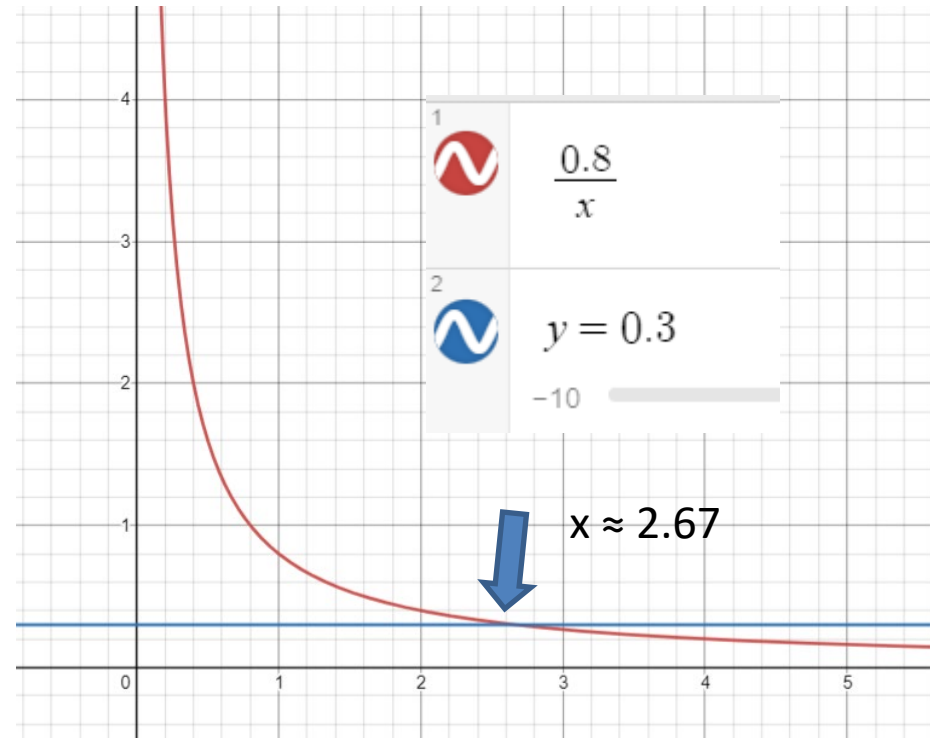
2.1 Roots of Equations [Method #1 - Graphical]

- When equation is linear combination of (at least) two functions
 - Roots located at intersecting points of curves
- Example
 - $f(x) = \frac{(0.8-0.3x)}{x}$
 - $f(x) = 0$
 - Rearrange $\rightarrow \frac{0.8}{x} = 0.3$

If $x = 2.67$

$$f(x) = \frac{(0.8 - 0.3(2.67))}{2.67}$$

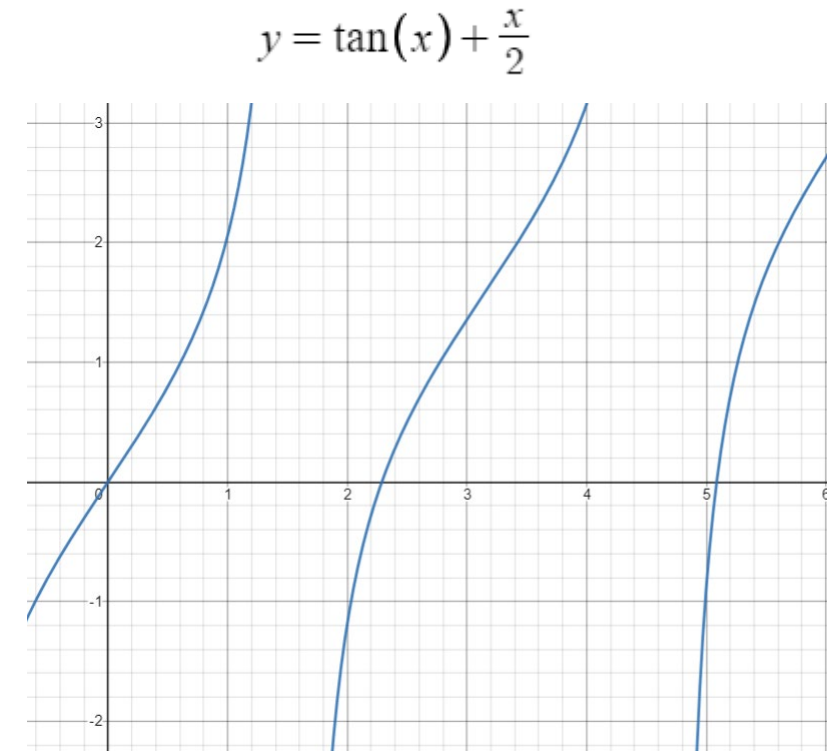
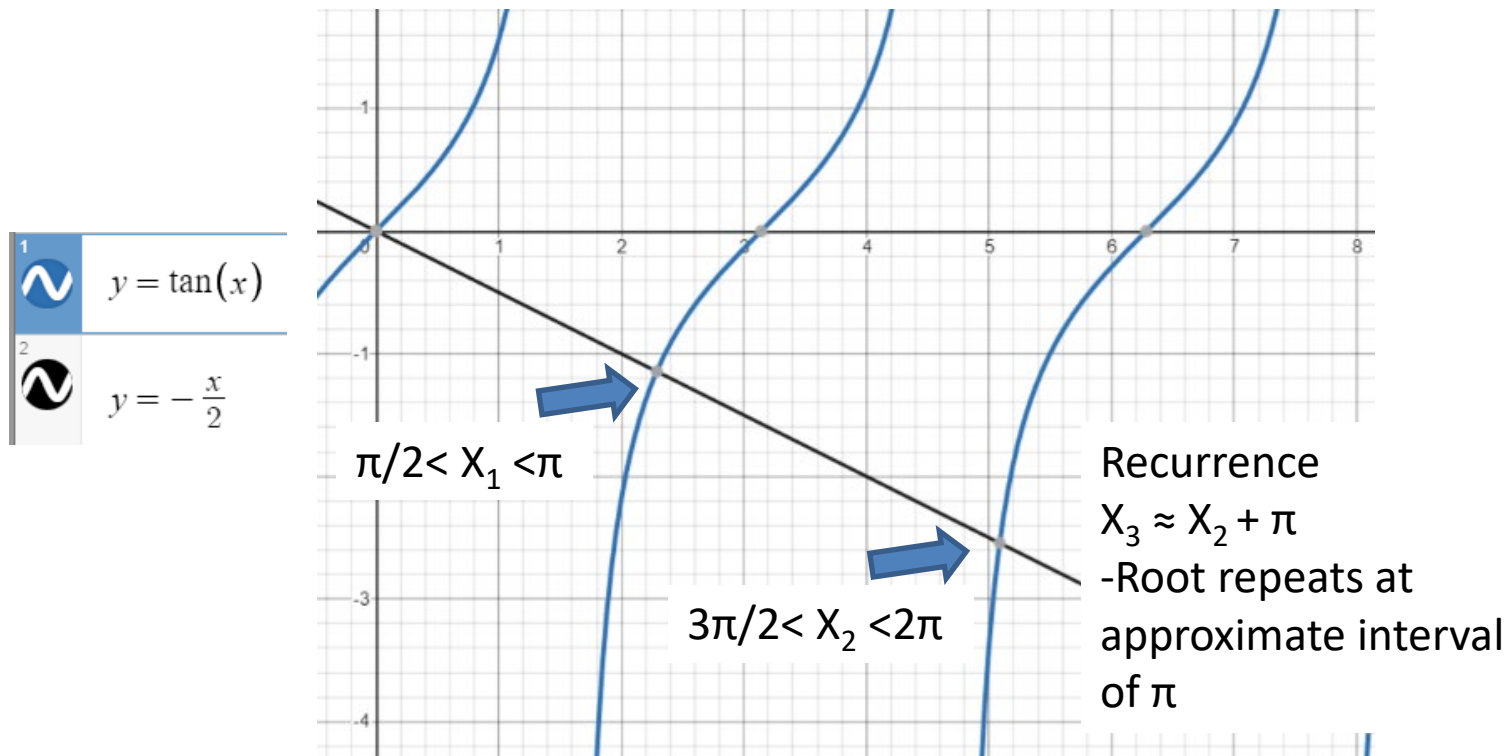
$$f(x) = 0$$



2.1 Roots of Equations [Method #1 - Graphical]

- Example

- Find all non-zero, positive roots of $\tan x = -\frac{x}{2}$



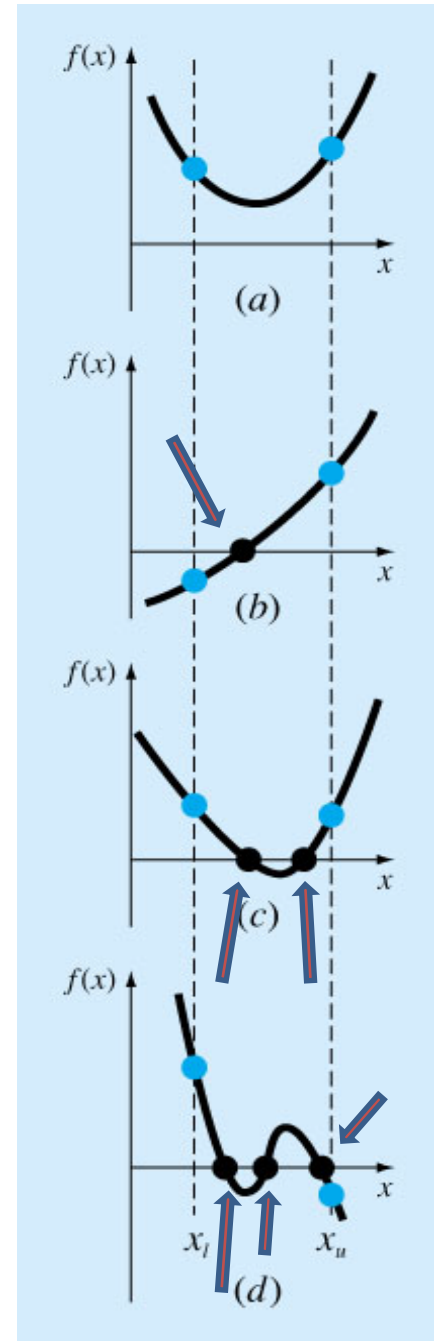
0,0
2.289,0
5.087,0
8.096,0
...

[Recall]

- Roots of equation
 - Values that satisfy homogeneous equation, solutions to implicit equation
- Graphical Methods
 - Graph function, locate roots where the curve crosses x axis -> predict number and approximate location
- Recall
 - General “rules of thumb” for root prediction based on graphical methods -> $f(x)$ changes sign on opposite side of the root

For today:

- Bracketing method [Bisection method]
 - Root located within interval prescribed by lower, upper bounds, prediction converges on true value of root
- Open methods [Fixed point iteration, Newton-Raphson, Secant]
 - Root located based on one or two starting values that do not necessarily bracket the root



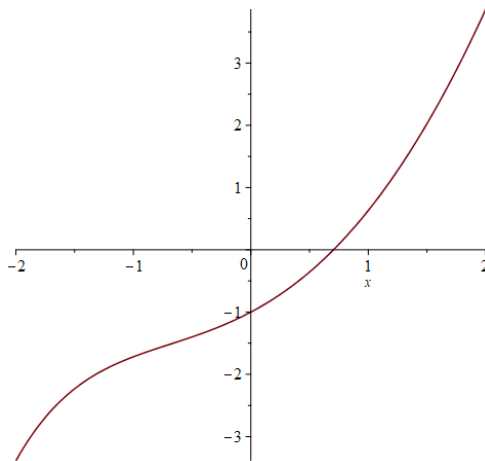
2.2 Bracketing Methods [Bisection Method]

- If $f(x)$ is real and continuous in the interval from x_l to x_u and $f(x_l)$ and $f(x_u)$ have opposite signs, i.e. $f(x_l) \cdot f(x_u) < 0$ then there is at least one real root between x_l and x_u

x_l — lower
 x_u — upper

- Incremental search methods
 - Locate and then divide an interval where the function changes sign -> one approach, bisection method divides intervals in half

- $f(x) = x^2 - e^{-x}$

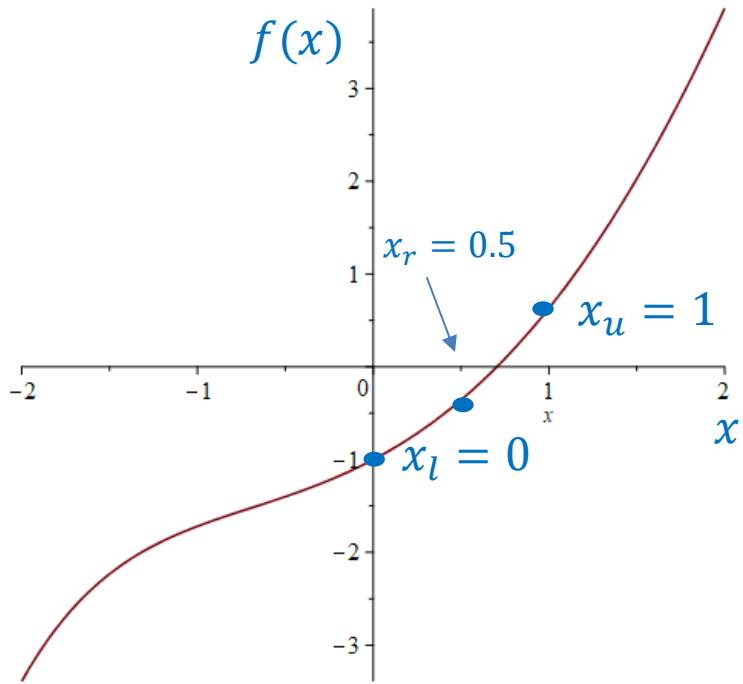


MatLab

```
x=-5:0.001:5;  
y=-exp(-x)+x.^2;  
hold on  
plot(x,y)  
grid on  
xlabel('x')  
ylabel('f(x)')  
scatter(fzero(@ (x) -exp(-x)+x.^2,0),0,'filled','red')  
hold off
```

2.2 Bisection Method

- $f(x) = x^2 - e^{-x}$



Step 1: Choose x_l and x_u such that function changes sign

Step 2: Estimate root by $x_r = \frac{x_l + x_u}{2}$

Step 3: Identify x_r position -> If $f(x_l)$ and $f(x_r)$ have the same sign,
then $x_l = x_r$
-> If $f(x_u)$ and $f(x_r)$ have the same sign,
then $x_u = x_r$

Step 4: Repeat steps 2 and 3

2.2 Bisection Method [Example]

- $f(x) = x^2 - e^{-x}$

iteration	x_l	x_u	x_r	$f(x_l)$	$f(x_u)$	$f(x_r)$	ε_a (%)
1	0	1	0.5	-1	0.632	-0.356	33%
2	0.5	1	0.75	-0.356	0.632	0.0901	17%
3	0.5	0.75	0.625	-0.356	0.0901	-0.145	9%
4	0.625	0.75	etc.	→			

- Continue until terminated based on specific criteria, e.g. specified approximate error

$$\varepsilon_a = \left[\frac{x_r^{new} - x_r^{old}}{x_r^{new}} \right] \cdot 100\%$$

Relative error Approximate

e.g. for the second iteration:

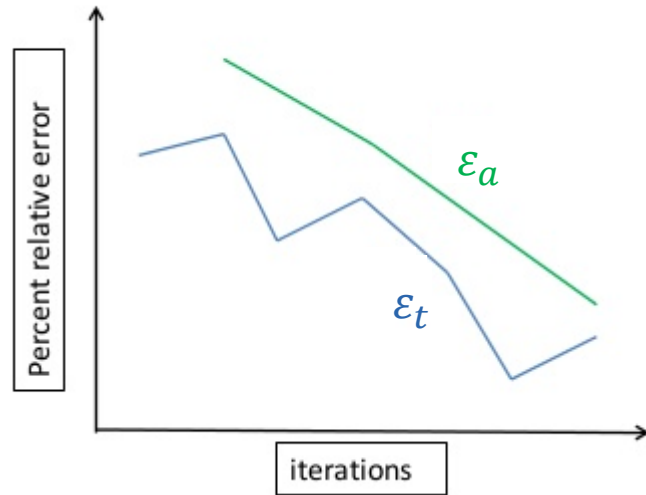
$$\varepsilon_a (\%) = \left[\frac{0.75 - 0.5}{0.75} \right] \cdot 100\% = 33\%$$

For the 10th iteration:

$$\varepsilon_a (\%) = 0.11\%$$

2.2 Bisection Method [Comparison of ε_a and ε_t]

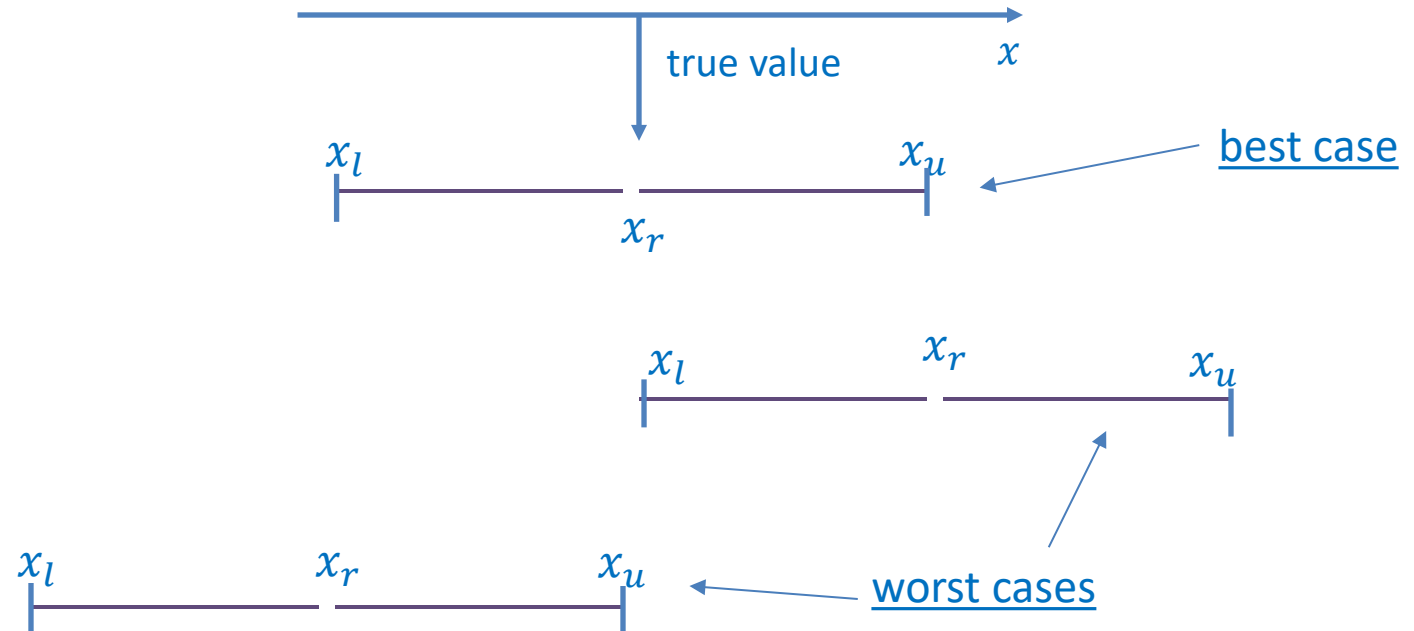
- Compare relative approximate error (ε_a) to relative true error (ε_t)



$\varepsilon_a > \varepsilon_t$ for the bisection method.

Why does ε_t increase and decrease?

The true root lies within $\pm \frac{\Delta x}{2}$



$$\left| \leftarrow \frac{\Delta x}{2} \rightarrow \right| \left| \leftarrow \frac{\Delta x}{2} \rightarrow \right|$$

$$x_r^{\text{new}} - x_r^{\text{old}} = \frac{\Delta x}{2} = \text{upper bound on } \varepsilon_t$$

(this is the bracketing interval)

2.2 Bisection Method [Error Level]

- Can calculate number of iterations to achieve desired error level

At start, iteration #0

Absolute error $\longrightarrow E_a^0 = \Delta x^0 = x_u^0 - x_l^0$

1st iteration $\longrightarrow E_a^1 = \frac{\Delta x^0}{2}$ $\longleftarrow$ Upper bound on error

2nd iteration $\longrightarrow E_a^2 = \frac{\Delta x^1}{2} = \frac{\Delta x^0}{2 \cdot 2}$

nth iteration $\longrightarrow E_a^n = \frac{\Delta x^0}{2^n}$ $\longleftarrow$ Solve for n

$$n = \frac{\log_{10}(\Delta x^0 / E_{a,d})}{\log_{10}(2)}$$

$E_{a,d}$ = desired absolute error level

If we know/set this value, we can determine the required number of iterations.

2.2 Bisection Method [Example]

- $f(x) = x^2 - e^{-x}$

$$E_{a,d} = 0.001$$

Starting values: $x_l = 0$ and $x_u = 1$

$$n = \frac{\log_{10}(\Delta x^0 / E_{a,d})}{\log_{10}(2)}$$

$$n = \frac{\log_{10}(1/0.001)}{\log_{10}(2)}$$

$$= 9.96$$

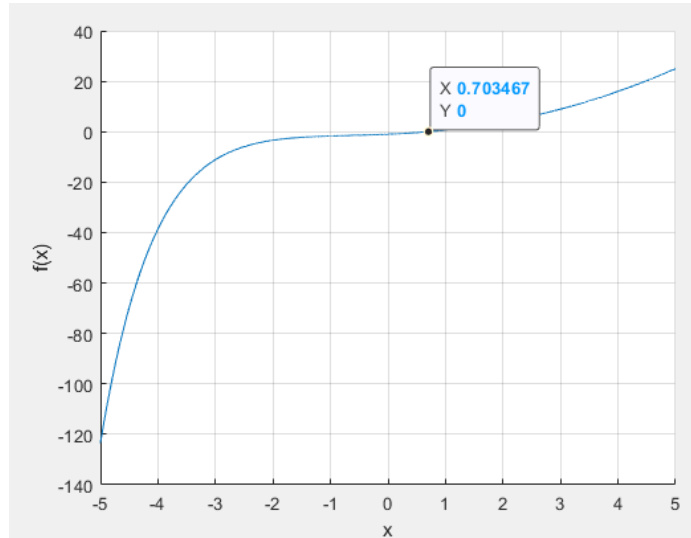
10 iterations to
exceed $E_{a,d} = 0.001$

Typically overestimates
based on upper limit error
value

2.2 Bisection Method [Example]

Iteration #	$x_{\text{low}} (x_l)$	$x_{\text{high}} (x_u)$	x_r	$f(x_l)$	$f(x_u)$	$f(x_r)$
0	0	1	0.5	-1	0.632121	-0.35653
1	0.5	1	0.75	-0.35653	0.632121	0.090133
2	0.5	0.75	0.625	-0.35653	0.090133	-0.14464
3	0.625	0.75	0.6875	-0.14464	0.090133	-0.03018
4	0.6875	0.75	0.71875	-0.03018	0.090133	0.02924
5	0.6875	0.71875	0.703125	-0.03018	0.02924	-0.00065
6	0.703125	0.71875	0.710938	-0.00065	0.02924	0.014249
7	0.703125	0.710938	0.707031	-0.00065	0.014249	0.006787
8	0.703125	0.707031	0.705078	-0.00065	0.006787	0.003065
9	0.703125	0.705078	0.704102	-0.00065	0.003065	0.001206
10	0.703125	0.704102	0.703613	-0.00065	0.001206	0.000277

Graphical (MatLab)



$$f(x) = x^2 - e^{-x}$$

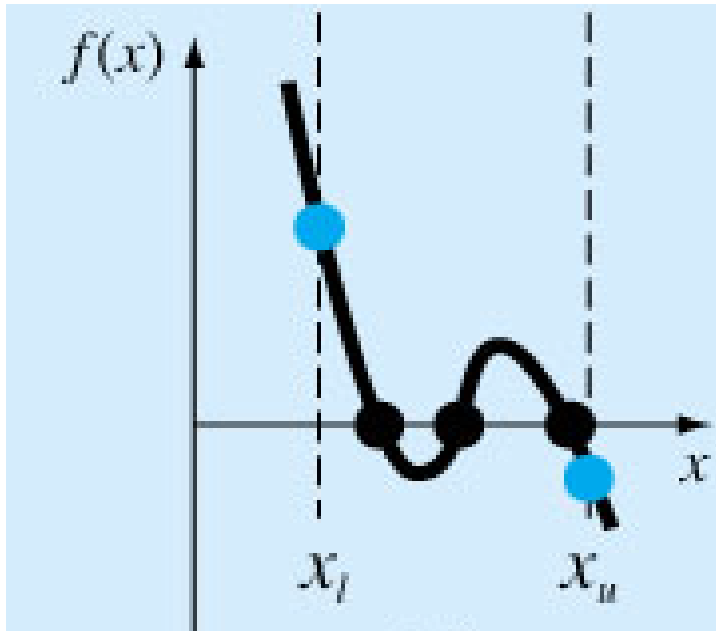
2.2 Bisection Method [Example]

Iteration #	x low (x_l)	x high (x_u)	x_r	$f(x_l)$	$f(x_u)$	$f(x_r)$
0	0	1	0.5	-1	0.632121	-0.35653
1	0.5	1	0.75	-0.35653	0.632121	0.090133
2	0.5	0.75	0.625	-0.35653	0.090133	-0.14464
3	0.625	0.75	0.6875	-0.14464	0.090133	-0.03018
4	0.6875	0.75	0.71875	-0.03018	0.090133	0.02924
5	0.6875	0.71875	0.703125	-0.03018	0.02924	-0.00065
6	0.703125	0.71875	0.710938	-0.00065	0.02924	0.014249
7	0.703125	0.710938	0.707031	-0.00065	0.014249	0.006787
8	0.703125	0.707031	0.705078	-0.00065	0.006787	0.003065
9	0.703125	0.705078	0.704102	-0.00065	0.003065	0.001206
10	0.703125	0.704102	0.703613	-0.00065	0.001206	0.000277

$$\begin{aligned}
 E_a &= |x_r^{new} - x_r^{old}| \\
 &= |0.704102 - 0.705078| \\
 &= 0.000976 < 0.001 \text{ after \#9}
 \end{aligned}$$

2.2 Bisection Method [Notes]

- Robust algorithm – always works
- Slow convergence – compared to other methods
- Cannot handle multiple roots ← graphing is important to show # of roots and locations



2.3 Open Methods

- Roots of equation
 - Solutions to implicit equation, values that satisfy homogeneous equation
- [recall] Bracketing methods
 - Root located within interval prescribed by lower, upper bounds, prediction converges on true value of root
- Open methods
 - Root located based on one or two starting values that do not necessarily bracket the root
 - A formula is used to predict the root
 - Methods include:
 - Fixed point iteration
 - Newton-Raphson
 - Secant method

2.3 Open Methods [Simple fixed-point iteration]

Given: $x = g(x_i)$

$x_{i+1} = g(x_i)$ ← Predict new value of x based on old value of x

Example: $x^2 - 2x + 3 = 0$

$$\text{Solve for } x = \frac{x^2+3}{2} \rightarrow x_{i+1} = \frac{x_i^2+3}{2}$$

Example: $\sin x + x^2 = 0$ ← Difficult to isolate x on LHS

Add x to both sides of equation

$$x_{i+1} = x_i + \sin x_i + x_i^2$$

Transform the equation to isolate 'x', or add 'x' to both sides so

$$f(x)=0$$

is transformed to

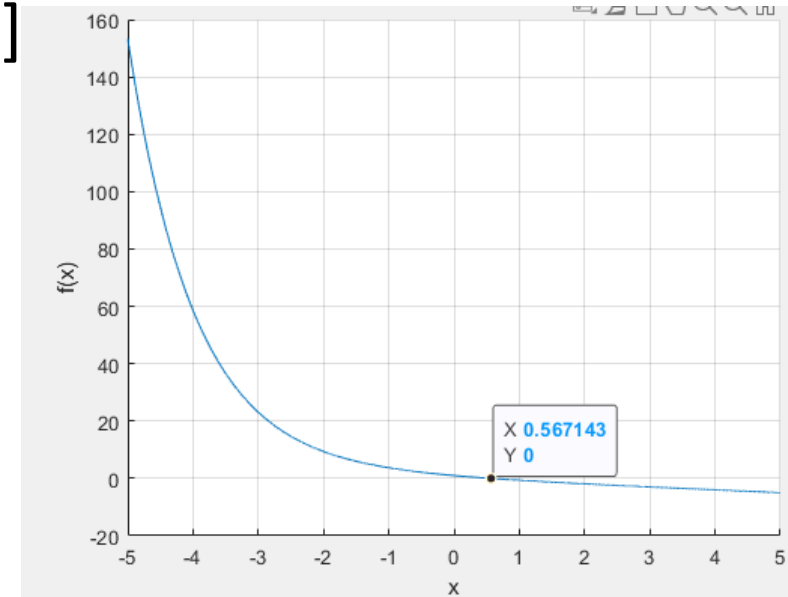
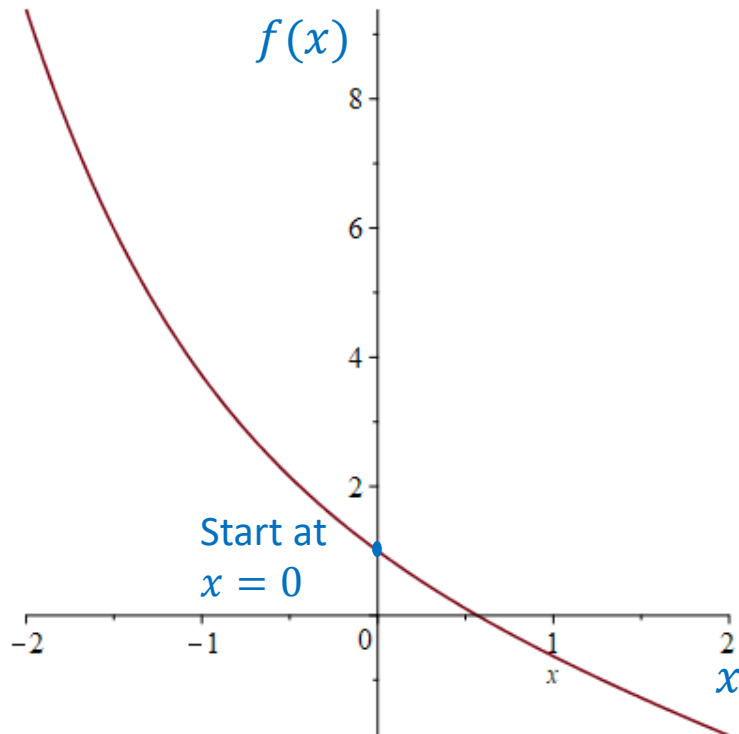
$$x=g(x)$$

* We also need an initial value to start the iterative process.

2.3 Open Methods [Simple fixed-point iteration - Example]

- $f(x) = e^{-x} - x = 0, x_{i+1} = e^{-x_i}$

$$\varepsilon_a = \left| \frac{x_{i+1} - x_i}{x_{i+1}} \right| \cdot 100\%$$



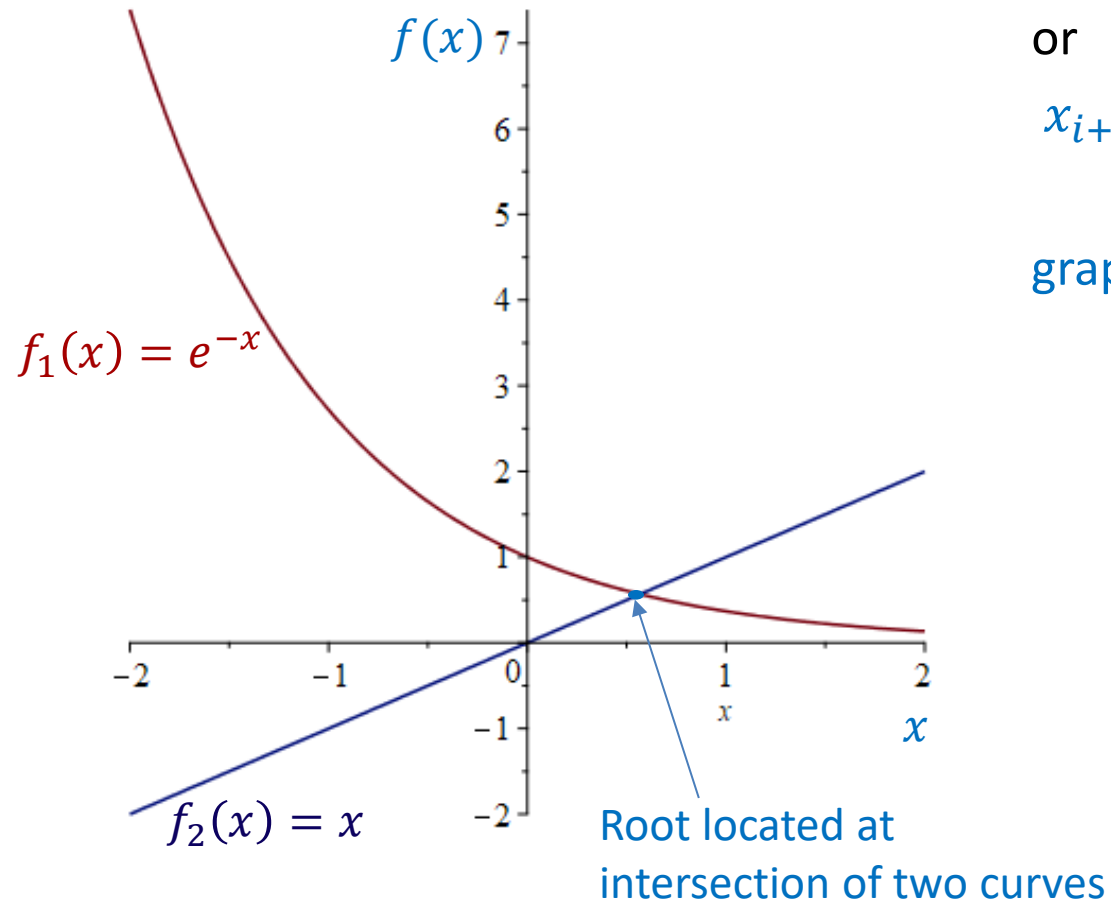
Iteration	x_i	x_{i+1}	ε_a (%)
1	0 (initial)	1	100
2	1	0.3679	172
3	0.3679	0.6922	47
...10	0.5711	0.5649	1.11

$$f(x) = e^{-x} - x = 0; x_{i+1} = g(x_i)$$

- Converges when $|g'(x)| < 1 \rightarrow$ slope at x
[see box 6.1 in Chapra for proof]

2.3 Open Methods [Two-Curve Graphical Example]

- What about different initial value? Does method always converge?



$$f(x) = e^{-x} - x = 0$$

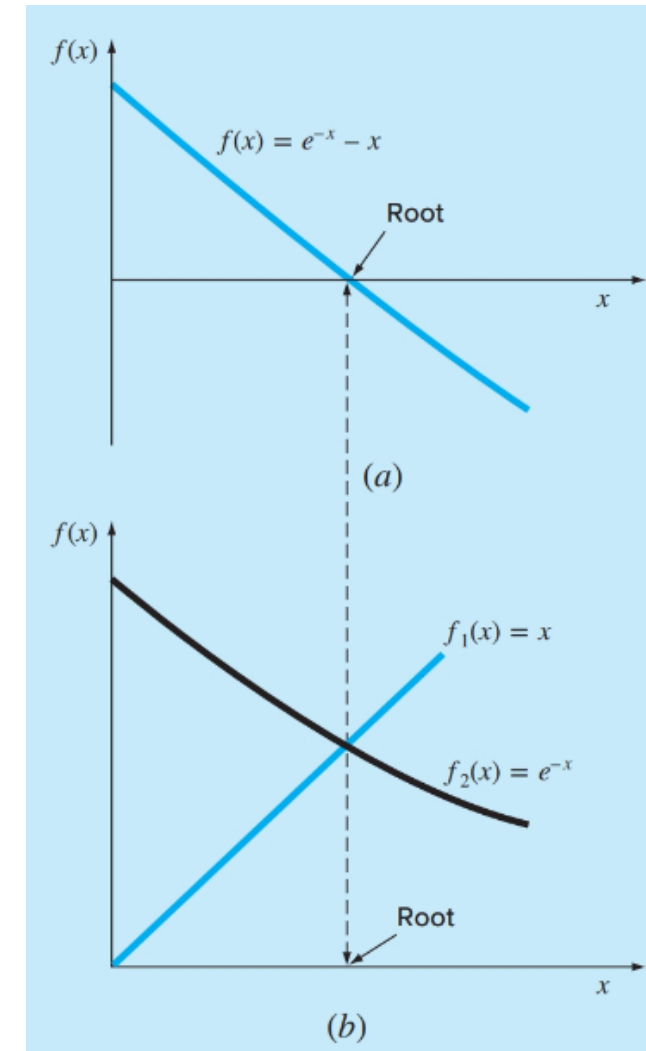
or

$$x_{i+1} = e^{-x_i}$$

graph as two functions

$$f_1(x) = e^{-x}$$

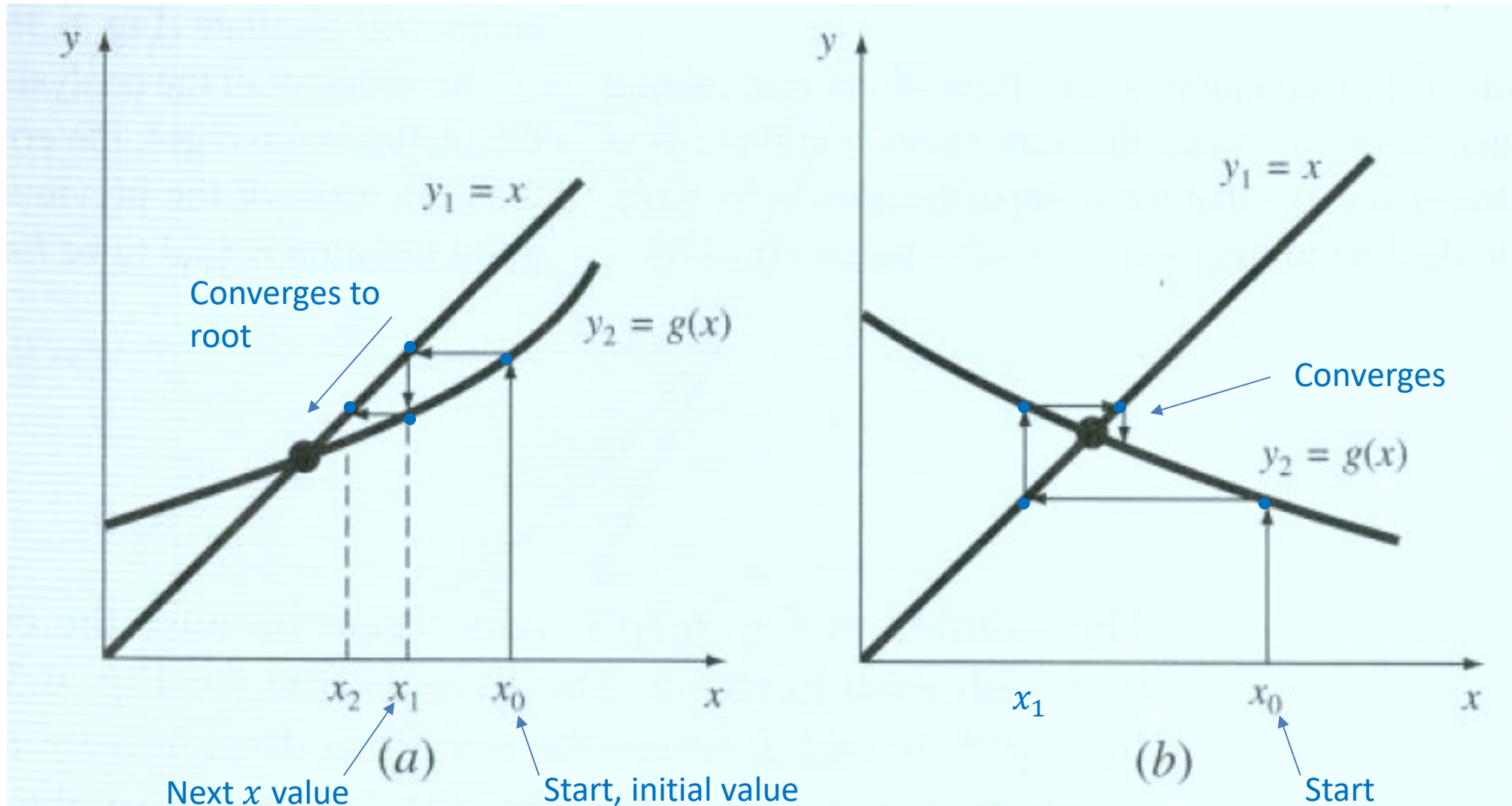
$$f_2(x) = x$$



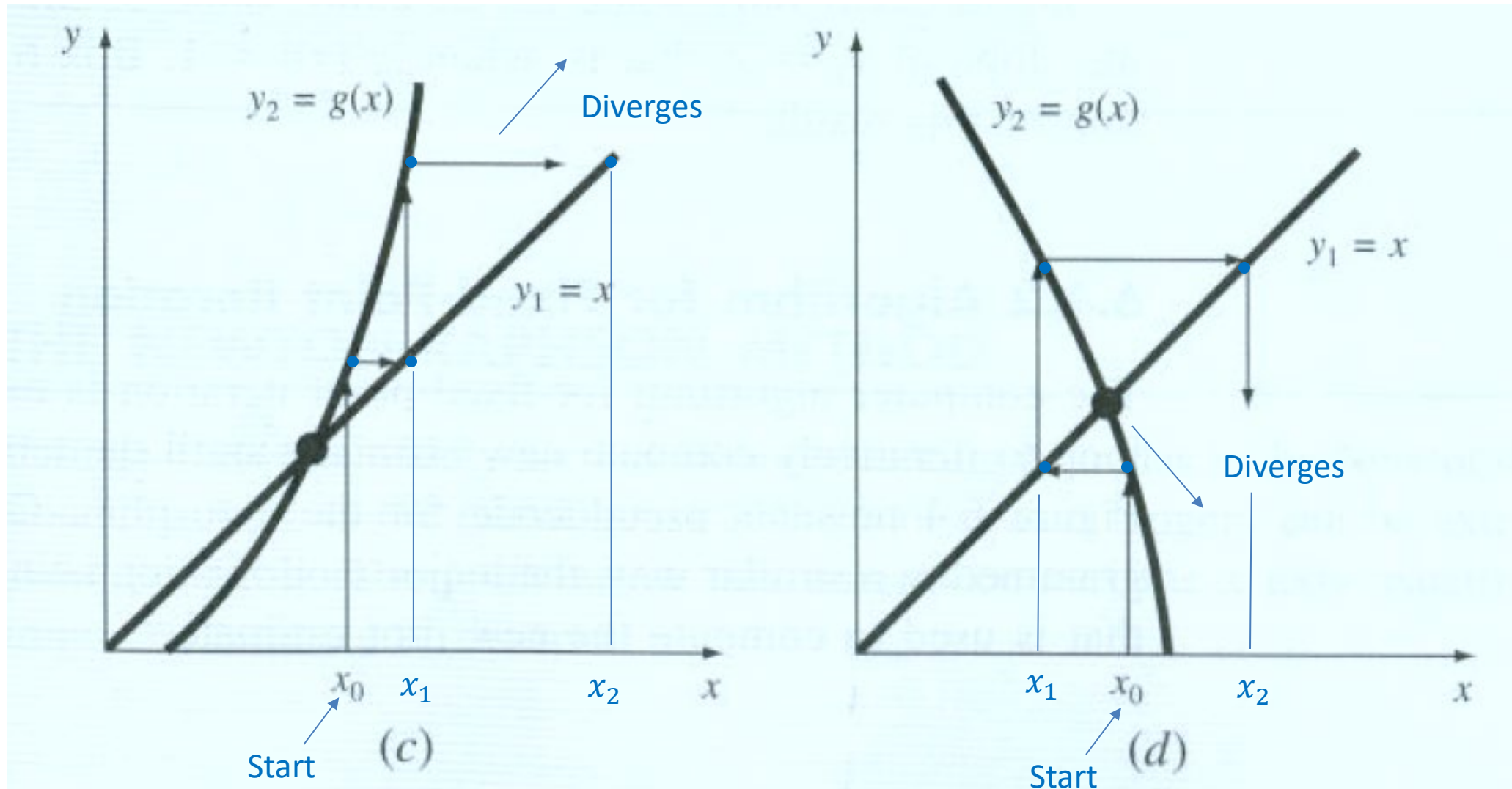
Section 6.2, Chapra

2.3 Open Methods [Converge]

[Iteration 'cobwebs']



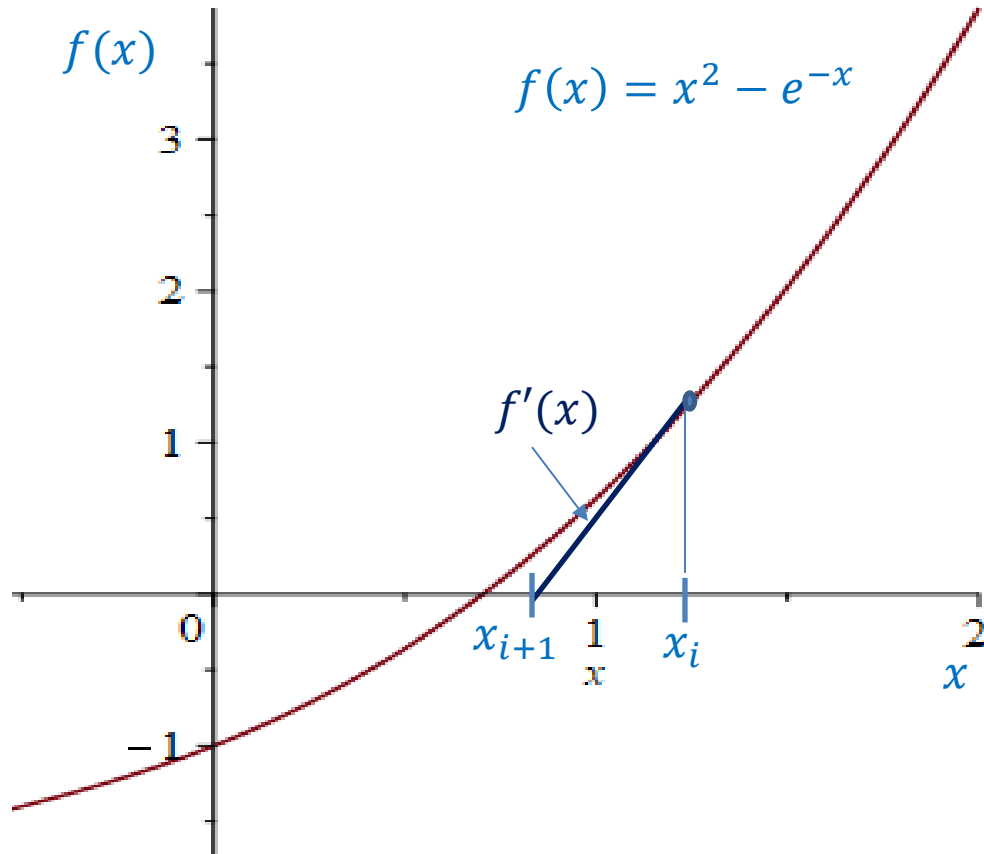
2.3 Open Methods [Diverge]



- Converges when $|g'(x)| < 1 \rightarrow$ slope at x
- Diverges when $|g'(x)| \geq 1$

2.4 Open Methods [Newton-Raphson]

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$



Step 1: Choose x_i near root

Step 2: Find slope of $f(x)$ at x_i , $f'(x_i)$

Step 3: Extend $f'(x_i)$ to x-axis to find x_{i+1}

$$f'(x_i) = \frac{\text{rise}}{\text{run}}$$

$$= \frac{f(x_i) - 0}{x_i - x_{i+1}}$$

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

Step 4: Repeat steps 2 and 3

2.4 Open Methods [Newton-Raphson - Example]

$$f(x) = x^2 - e^{-x}$$

$$f'(x) = 2x + e^{-x}$$

$$x_{i+1} = x_i - \frac{x_i^2 - e^{-x_i}}{2x_i + e^{-x_i}}$$

iteration	x_i	f/f'	x_{i+1}
1	1 (initial)	$\frac{1^2 - e^{-1}}{2(1) + e^{-1}}$	0.73304
2	0.73304	$\frac{x_i^2 - e^{-x}}{2x_i + e^{-x}}$	0.70381
3	0.70381	$\frac{x_i^2 - e^{-x}}{2x_i + e^{-x}}$	0.7034675
4	0.7034675	$\frac{x_i^2 - e^{-x}}{2x_i + e^{-x}}$	0.7034674

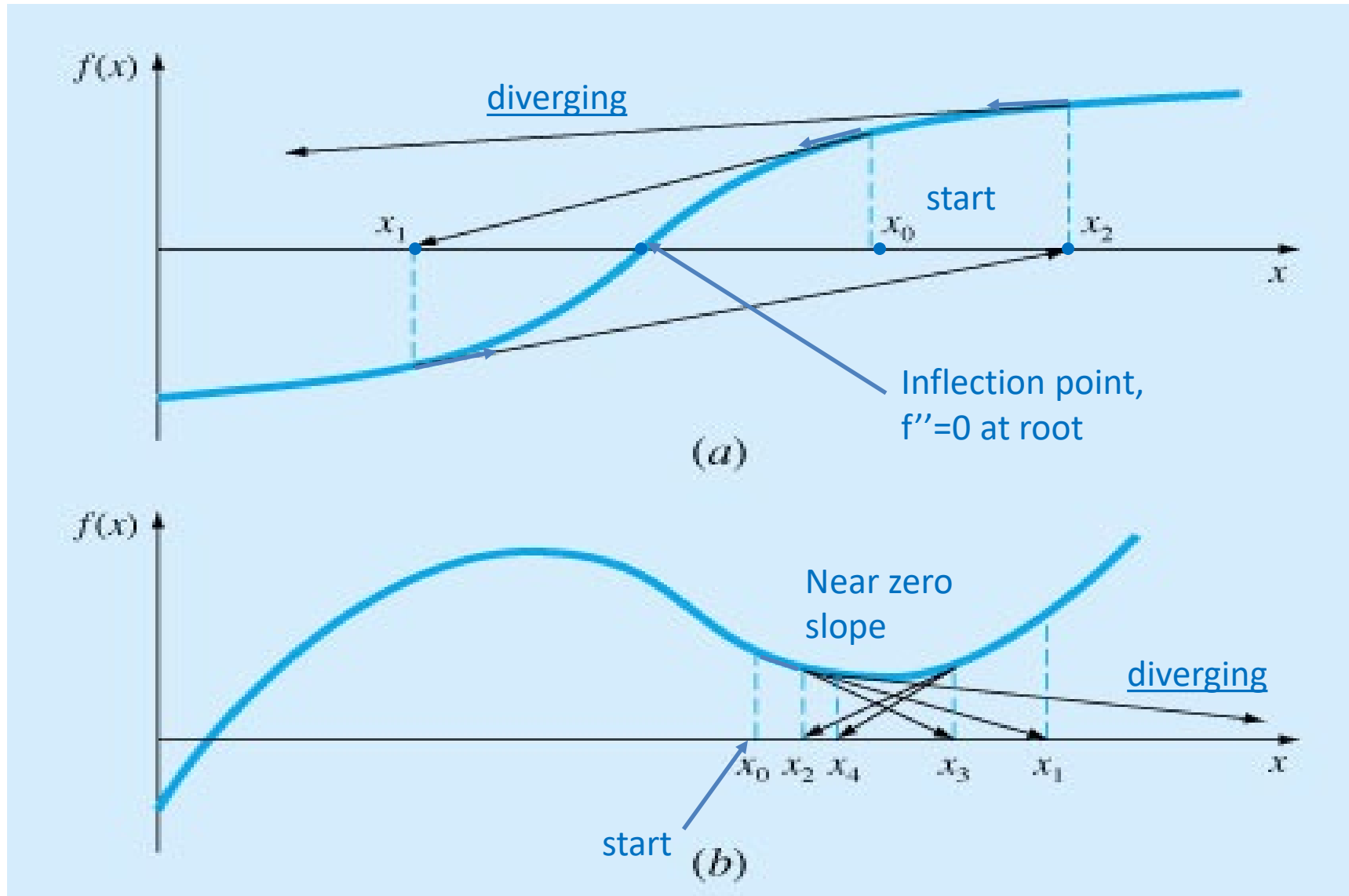
2.4 Open Methods [Newton-Raphson]

- Approximate relative error
 - $\varepsilon_a = \left| \frac{x_{i+1} - x_i}{x_{i+1}} \right| \cdot 100\%$ <- to monitor convergence
- Absolute true error
 - $E_{t,i+1} \propto (E_{t,i})^2$ <- second order method
- Cannot predict #iterations to convergence
 - Why? Function is initial value dependent.

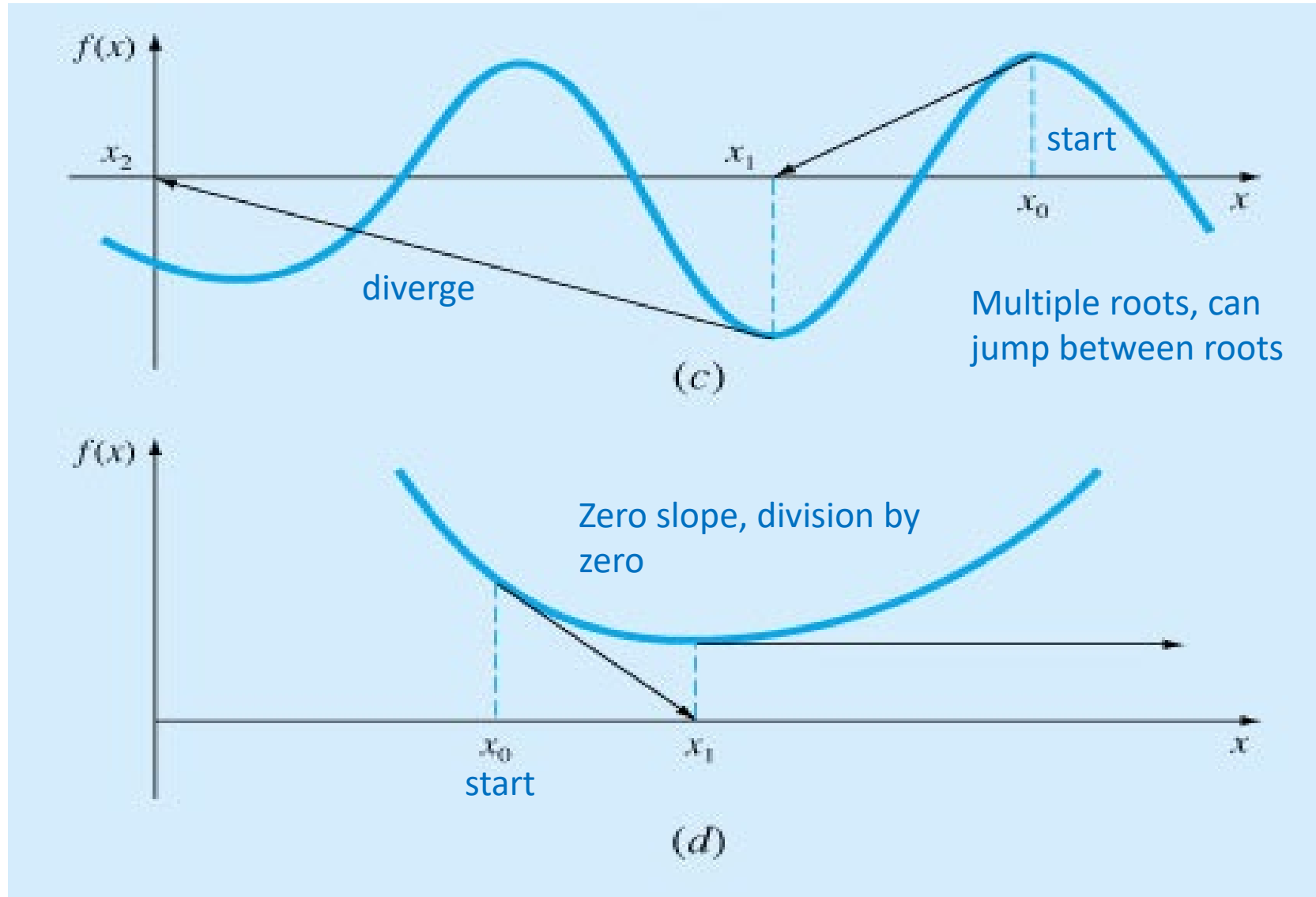
The error is approximately proportional to the square of the previous error (see box 6.2 in text for proof).

The number of correct decimal places (approx.) doubles for each iteration – known as quadratic convergence.

2.4 Open Methods [Potential Challenges for Newton-Raphson]



2.4 Open Methods [Potential Challenges for Newton-Raphson]



2.4 Open Methods [Newton-Raphson]

- Roots of equation
 - Solutions to implicit equation, values that satisfy homogeneous equation
- Recall, for Newton-Raphson
 - Open method, root located based on one starting value
$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$
 - Where $f'(x_i)$ = slope of curve at x_i , follow tangent line to x axis, repeat until converged

In general:

- Faster convergence (compared to other methods)
- Simple algorithm compared to bisection (no if-else statements required)
- Cannot handle multiple roots, can diverge if initial value not chosen properly
- One challenge is that Newton-Raphson requires evaluation of the function derivative
 - Polynomials, trigonometry, powers, exponentials, etc. -> [easy](#)
 - Other functions difficult or impossible to evaluate

2.4 Open Methods [Newton-Raphson]

- Recommendations
 1. Always plot the function (include a plotting routine in your program) to check/monitor the solution.
 2. When the termination criterion is achieved, always substitute the calculated root back into the original equation to determine if the result is close to zero (guards against slow convergence or oscillation, where the approximate error may be small but the solution is far from the root).
 3. Include an upper limit on the number of iterations (for slow convergence, oscillating solutions, or divergent solutions).
 4. Check to make sure the function derivative does not equal zero, if $f'(x)=0$ then f/f' is a divide by zero.

2.5 Secant Method

The Secant Method uses an approximation of the derivative (backward difference) in the equation:

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

- Similar to Newton-Raphson but approximate derivative using backward difference
- Backward difference
 - First order (linear) approximation of derivative using previous value

$$f'(x_i) \cong \frac{f(x_{i-1}) - f(x_i)}{x_{i-1} - x_i}$$

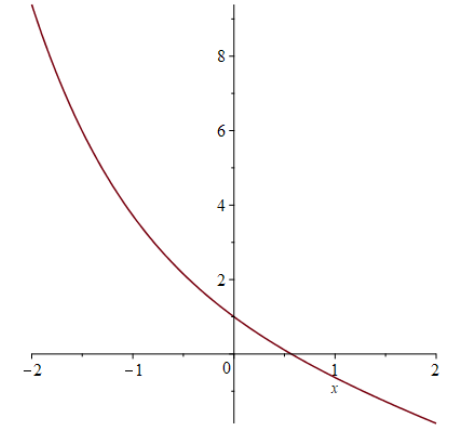
$$x_{i+1} = x_i - \frac{f(x_i)(x_{i-1} - x_i)}{f(x_{i-1}) - f(x_i)}$$

Sub into Newton-Raphson formula

2.5 Secant Method [Example]

- $f(x) = e^{-x} - x$
$$x_{i+1} = x_i - \frac{f(x_i)(x_{i-1} - x_i)}{f(x_{i-1}) - f(x_i)}$$
- Requires two starting points *

i	x_{i-1}	x_i	$f(x_{i-1})$	$f(x_i)$	x_{i+1}	ε_a (%)
1	0 *	1 *	1	-0.63212	0.612699837	8.66586
2	1	0.6127	-0.63212	-0.07081	0.563838389	0.587472
3	0.6127	0.563838	-0.07081	0.005182	0.567170358	0.00477
4	0.563838	0.56717	0.005182	-4.2E-05	0.567143307	2.9E-06
5	0.56717	0.567143	-4.2E-05	-2.5E-08	0.56714329	



Similar convergence to
Newton-Raphson, but
didn't have to find f'

2.5 Application Example

- Fluid flow in a pipe
- Colebrook equation
 - Turbulent (high speed) flow

$$\frac{1}{\sqrt{f}} + 2.0 \log \left(\frac{\varepsilon}{3.7 D} + \frac{2.51}{Re \sqrt{f}} \right) = 0$$

- Where;
 - f = friction factor, resistance to flow
 - ε = roughness (m)
 - D = diameter (m)
 - Re = Reynold's number = $\frac{\rho V D}{\mu}$ = dimensionless velocity

Equation used to calculate friction coefficients for fluid flows

ε / D = relative roughness

μ = dynamic viscosity

V = fluid velocity

ρ = density

2.5 Application Example

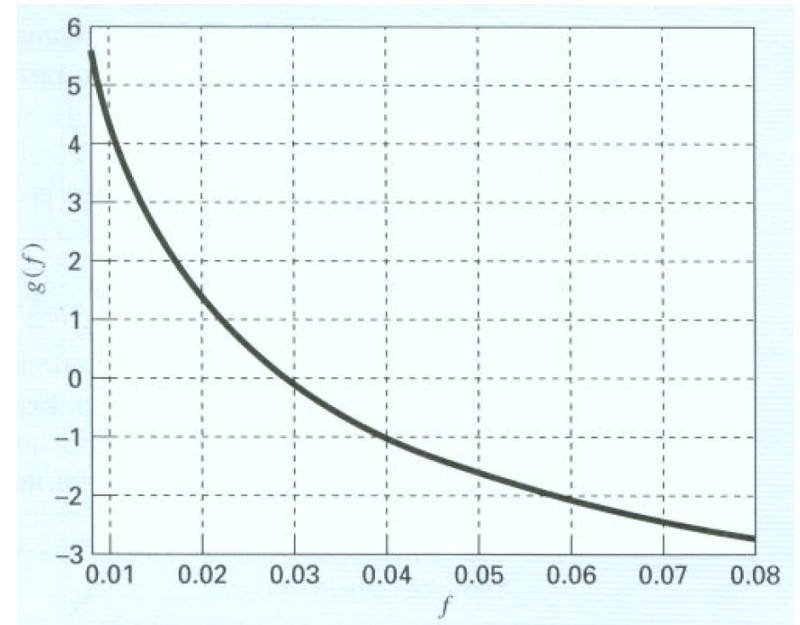
$$g(f) = \frac{1}{\sqrt{f}} + 2.0 \log \left(\frac{\varepsilon}{3.7 D} + \frac{2.51}{Re \sqrt{f}} \right) = 0$$

- Given

- $\rho = 1.23 \text{ kg/m}^3$
- $\mu = 1.79 \times 10^{-5} \text{ N} \cdot \text{s/m}^2$
- $D = 0.005 \text{ m}$
- $V = 40 \text{ m/s}$
- $\varepsilon = 0.0015 \text{ mm}$

Solve for f knowing that
 $0.008 < f < 0.08$
(empirical data)
(upper and lower bounds)

$$g(f) = \frac{1}{\sqrt{f}} + 2.0 \log \left(\frac{0.0015 \times 10^{-3}}{3.7 (0.005)} + \frac{2.51}{13,743 \sqrt{f}} \right)$$



2.5 Application Example - Bisection Method

- $$g(f) = \frac{1}{\sqrt{f}} + 2.0 \log \left(\frac{0.0015 \times 10^{-3}}{3.7 (0.005)} + \frac{2.51}{13,743 \sqrt{f}} \right)$$

i	f_l	f_u	f_r	$g(f_l)$	$g(f_u)$	$g(f_r)$	ε_a
0	0.008	0.08	0.044	5.834	-2.741	-1.275	
1	0.008	0.044	0.026	5.834	-1.275	0.37	69.23
20	0.028968	0.028968	0.028968	7.86E-06	-5.8E-08	3.9E-06	0.000119
21	0.028968	0.028968	0.028968	3.9E-06	-5.8E-08	1.92E-06	5.93E-05

Root = friction
factor (f)

ε_a

2.5 Application Example - Newton-Raphson

$$g(f) = \frac{1}{\sqrt{f}} + 2.0 \log \left(\frac{\varepsilon}{3.7 D} + \frac{2.51}{Re \sqrt{f}} \right) = 0$$

Evaluation of the derivative is challenging!

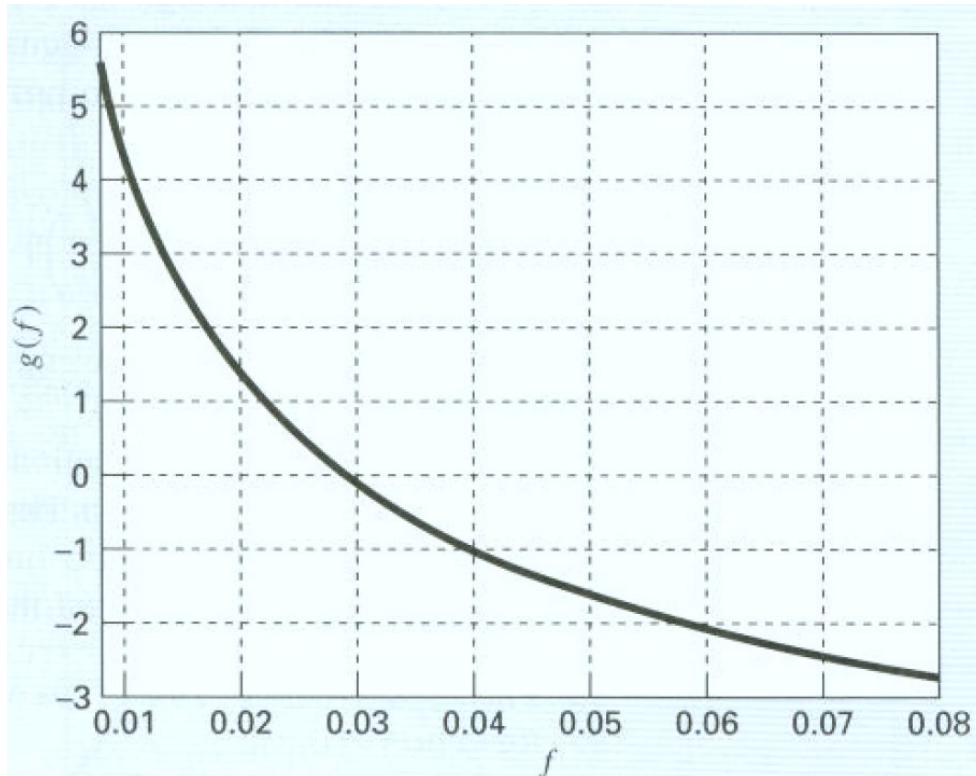
$$\frac{g(f_i)}{g'(f_i)}$$



$$\begin{aligned} &g := \frac{1}{\text{sqrt}(f)} + 2 \cdot \log_{10} \left(\frac{0.0015e-3}{(3.7 \cdot 0.005)} + \frac{2.51}{(13743 \cdot \text{sqrt}(f))} \right); \\ &g := \frac{1}{\sqrt{f}} + \frac{2 \ln \left(0.00008108108108 + \frac{0.0001826384341}{\sqrt{f}} \right)}{\ln(10)} \\ &\text{diff}(g, f); \\ &\quad -\frac{1}{2f^{3/2}} - \frac{0.0001826384341}{f^{3/2} \left(0.00008108108108 + \frac{0.0001826384341}{\sqrt{f}} \right) \ln(10)} \\ &\text{simplify} \left(\frac{g}{\text{diff}(g, f)} \right); \\ &\quad \frac{0.00008108108108 \left(2 \cdot \ln \left(\frac{0.00008108108108 (\sqrt{f} + 2.252540687)}{\sqrt{f}} \right) \sqrt{f} + \ln(2) + \ln(5) \right) f (\sqrt{f} + 2.252540687)}{0.0002759864784 \sqrt{f} + 0.0002102702678} \end{aligned}$$

2.5 Application Example - Newton-Raphson

iteration	f	g/g'	f_{new}	ε_a
1	0.008	-0.00777	0.015769	49.26693
2	0.015769	-0.00838	0.024154	34.71444
3	0.024154	-0.00422	0.02837	14.86287
4	0.02837	-0.00059	0.028959	2.032748
5	0.028959	-8.9E-06	0.028968	0.030841
6	0.028968	-2E-09	0.028968	6.87E-06



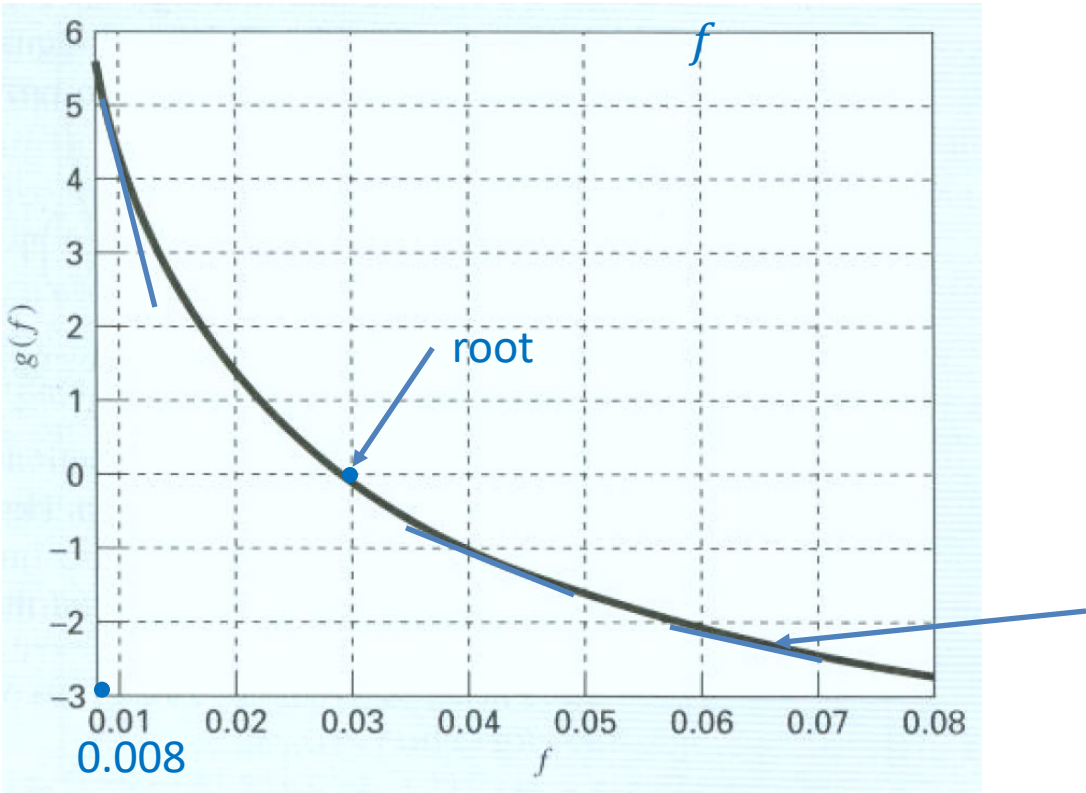
iteration	f	g/g'	f_{new}	ε_a
1	0.05	0.030812	0.019188	160.5862
2	0.019188	-0.00722	0.026403	27.32794
3	0.026403	-0.0024	0.028801	8.326348
4	0.028801	-0.00017	0.028967	0.573727
5	0.028967	-6.9E-07	0.028968	0.002399
6	0.028968	-1.2E-11	0.028968	4.15E-08

iteration	f	g/g'	f_{new}	ε_a
1	0.07	0.075141	-0.00514	1461.581
2	-5.14E-03	#NUM!	#NUM!	#NUM!
3	#NUM!	#NUM!	#NUM!	#NUM!
4	#NUM!	#NUM!	#NUM!	#NUM!
5	#NUM!	#NUM!	#NUM!	#NUM!
6	#NUM!	#NUM!	#NUM!	#NUM!

2.5 Application Example - Newton-Raphson

iteration	f	g/g'	f_{new}	ϵ_a
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6	0.028968	-1.2E-11	0.028968	4.15E-08



iteration	f	g/g'	f_{new}	ϵ_a
1	0.07	0.075141	-0.00514	1461.581
2	-5.14E-03	#NUM!	#NUM!	#NUM!
3	#NUM!	#NUM!	#NUM!	#NUM!
4	#NUM!	#NUM!	#NUM!	#NUM!
5	#NUM!	#NUM!	#NUM!	#NUM!
6	#NUM!	#NUM!	#NUM!	#NUM!

Solution diverges!

For $f_1 > 0.066$, solution won't converge

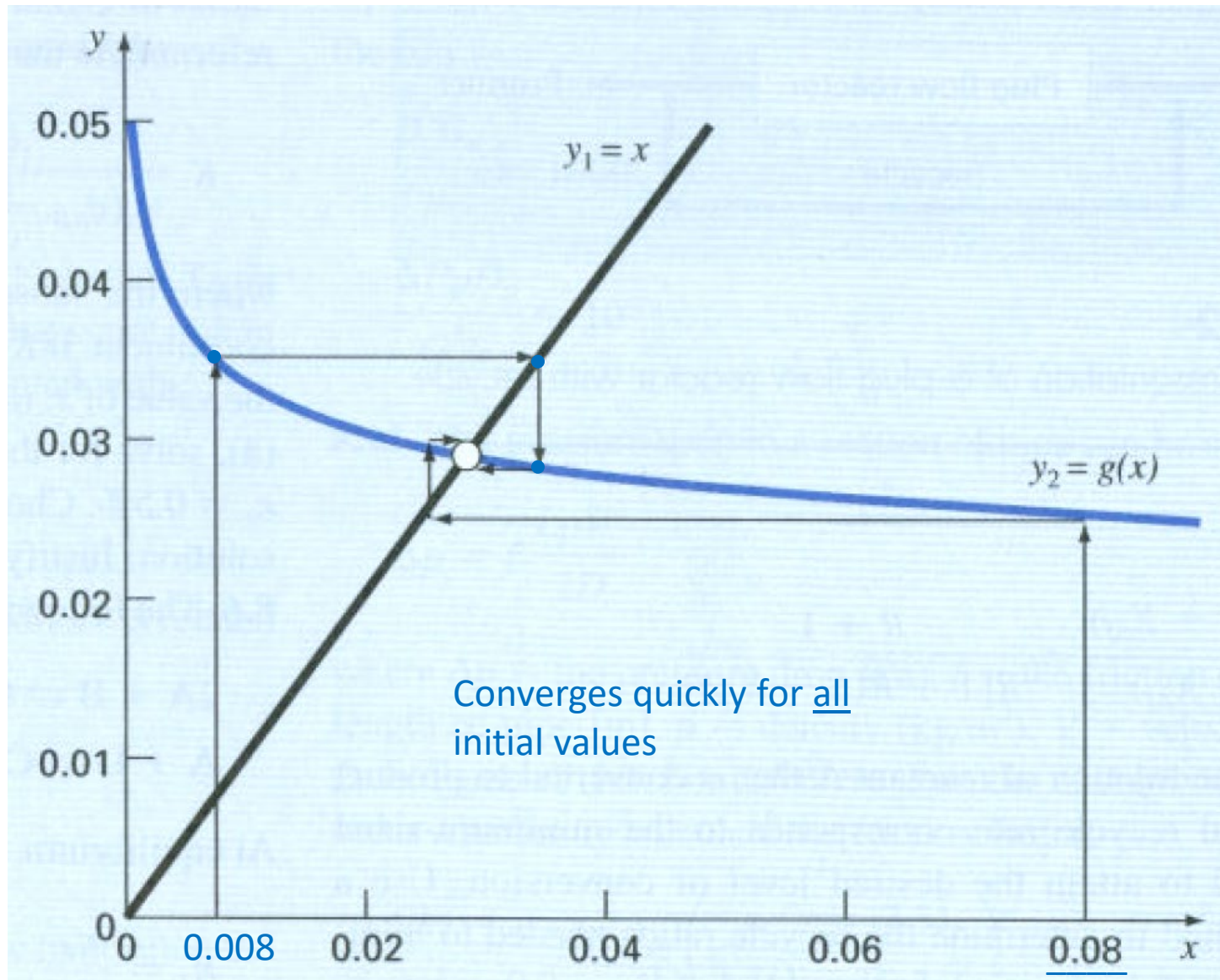
2.5 Application Example - Fixed Point Iteration

$$g(f) = \frac{1}{\sqrt{f}} + 2.0 \log \left(\frac{0.0015 \times 10^{-3}}{3.7 (0.005)} + \frac{2.51}{13,743 \sqrt{f}} \right) = 0$$



$$f_{i+1} = \frac{0.25}{\left[\log \left(\frac{0.0015 \times 10^{-3}}{3.7(0.005)} + \frac{2.51}{13743\sqrt{f_i}} \right) \right]^2}$$

2.5 Application Example - Fixed Point Iteration



In this case, simple fixed-point iteration works well and converges quickly.

Note: one single method is not always best for all problems.

Important to identify when methods break down, and which method may work best for a particular problem.

2.6 Summary – Roots of Eqn

TABLE PT2.4 Summary of important information presented in [Part Two](#).

Method	Formulation	Graphical Interpretation	Errors and Stopping Criteria
Bracketing methods:			
Bisection	$X_r = \frac{X_l + X_u}{2}$ If $f(X_l)f(X_r) < 0$, $X_u = X_r$ $f(X_l)f(X_r) > 0$, $X_l = X_r$		Stopping criterion: $\left \frac{X_r^{\text{new}} - X_r^{\text{old}}}{X_r^{\text{new}}} \right 100\% \leq \epsilon_s$
False position	$X_r = X_u - \frac{f(X_u)(X_l - X_u)}{f(X_l) - f(X_u)}$ If $f(X_l)f(X_r) < 0$, $X_u = X_r$ $f(X_l)f(X_r) > 0$, $X_l = X_r$		Stopping criterion: $\left \frac{X_r^{\text{new}} - X_r^{\text{old}}}{X_r^{\text{new}}} \right 100\% \leq \epsilon_s$
Newton-Raphson	$X_{i+1} = X_i - \frac{f(X_i)}{f'(X_i)}$		Stopping criterion: $\left \frac{X_{i+1} - X_i}{X_{i+1}} \right 100\% \leq \epsilon_s$ Error: $E_{i+1} = O(E_i^2)$
Secant	$X_{i+1} = X_i - \frac{f(X_i)(X_{i-1} - X_i)}{f(X_{i-1}) - f(X_i)}$		Stopping criterion: $\left \frac{X_{i+1} - X_i}{X_{i+1}} \right 100\% \leq \epsilon_s$

We didn't cover this method

